

Service Analysis with Openrouteservice

Amandine Holemans



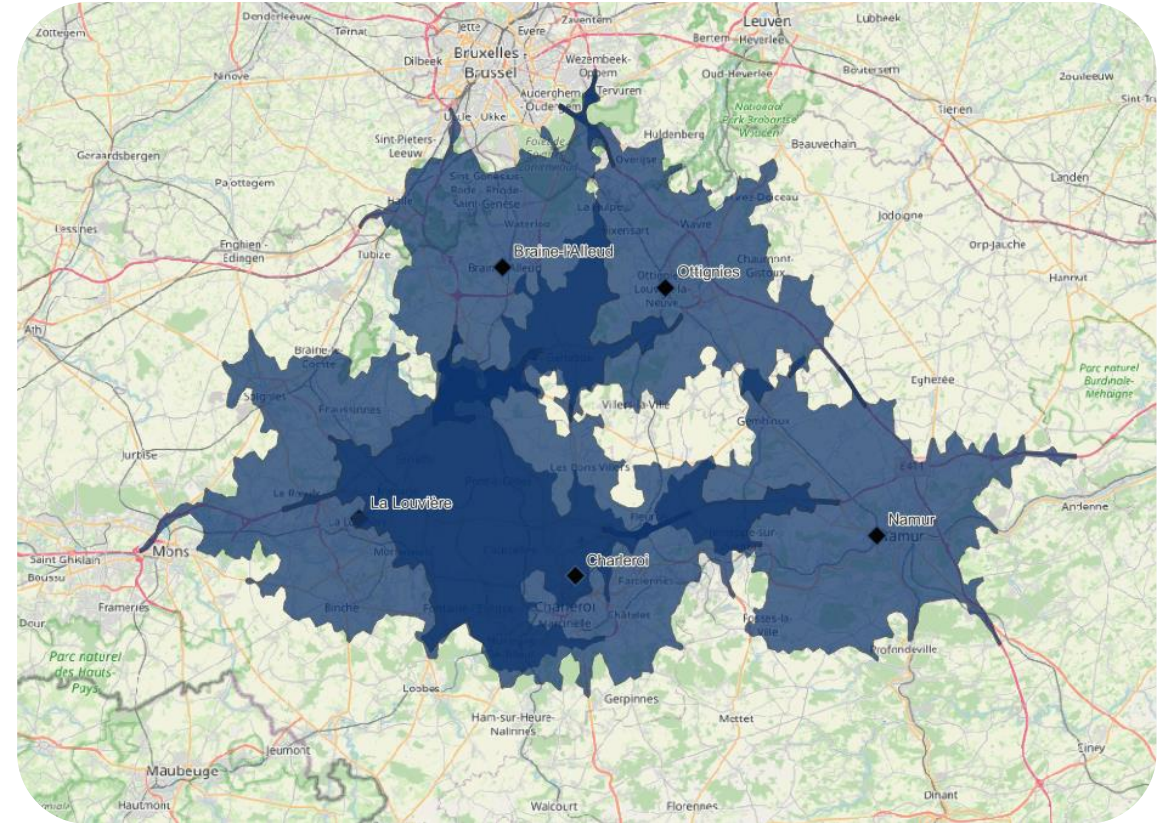
Smart Geo Insights



www.gim.be

Agenda

1. Description of the project
2. Description of the tools
 1. Openroadservice
 2. QGIS and plugin
 3. Dynamic temporal control
3. Step by step
4. Conclusion





Analysis of Emergency service locations

In collaboration with SPI (Liège)



Purpose: Analyse the optimal configuration for the location of emergency services



Use of the Qgis project: Analyse, present and convince



Challenges:

Different types of required response time
Different types of possible scenario to present
Presentation made by nonuser of Qgis



Presentation:

Use of fictive data
Could be applied to different types of services



Openroadservice



Openroadservice (ORS)

Description



A **web API** providing access to **road network data** and routing capabilities.

- Powered by **OSM data**
- Delivers **road information** and metadata
- Supports **route calculation** and spatial queries
- Based on **open standards** for easy integration (REST / OGC)
- Enables **smarter mobility**

Source: <https://openrouteservice.org/services/>

Openroad Service (ORS)

Types of services



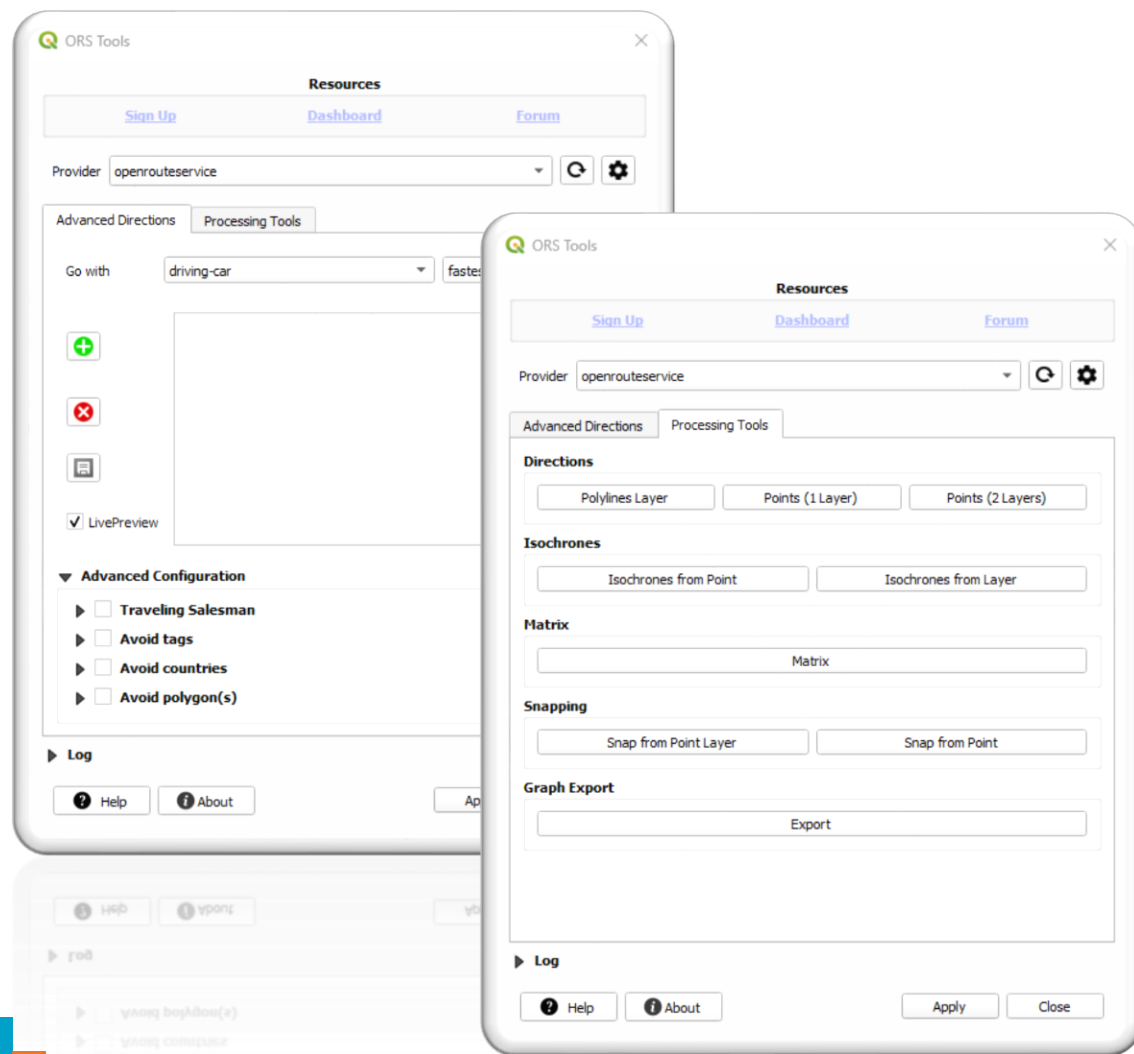
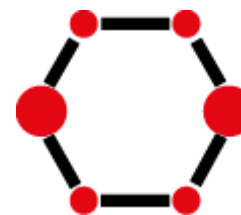
- Isochrones
- Direction
- Optimization
- Time-distance Matrix
- Geocoding
- Poi
- Elevation
- ...

Source: <https://openrouteservice.org/services/>

Qgis and plugin



QGIS and ORS plugin



- **Easy** to use, no API knowledge needed
- Integrates **all ORS tools**
- **Free sign up** required
- **Quotas** by minutes and by days linked to the user (can be upgraded)

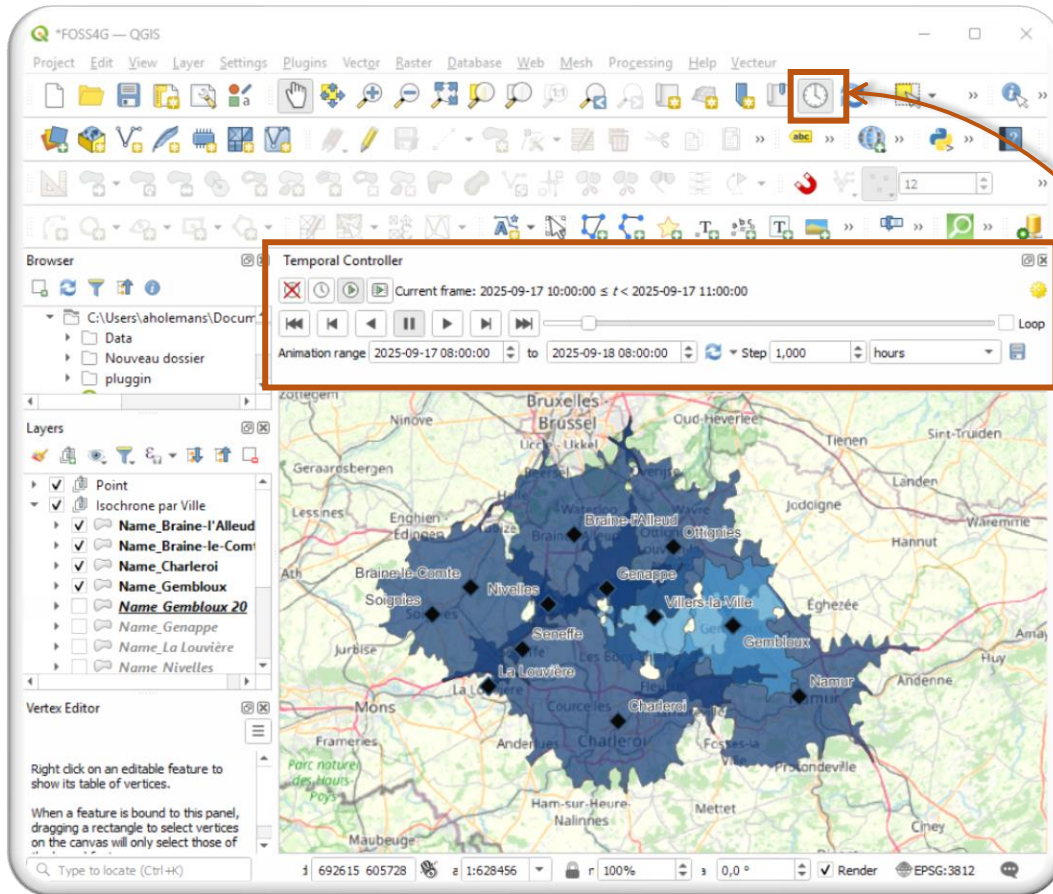
Main Endpoints	Total Quota Left	Quota per Minute
Directions	2000	40
Export	100	5
Isochrones	500	20
Matrix	500	40
Snap	2000	100

<https://account.heigit.org/manage/key>

Dynamic temporal control

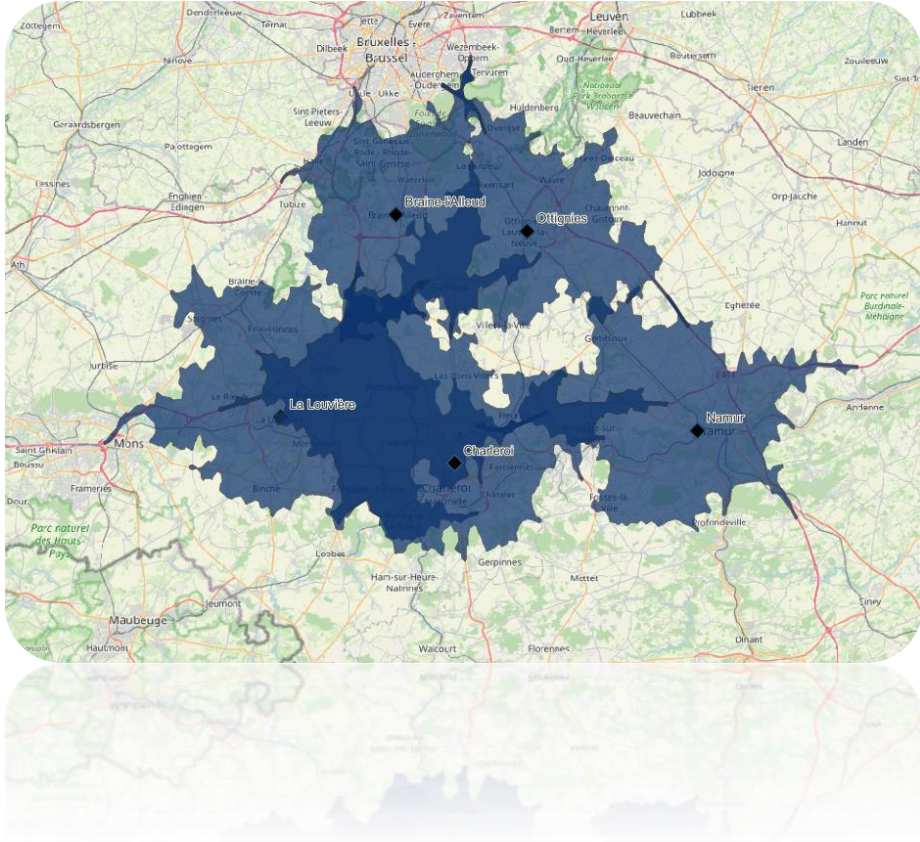


Presentation of the tool



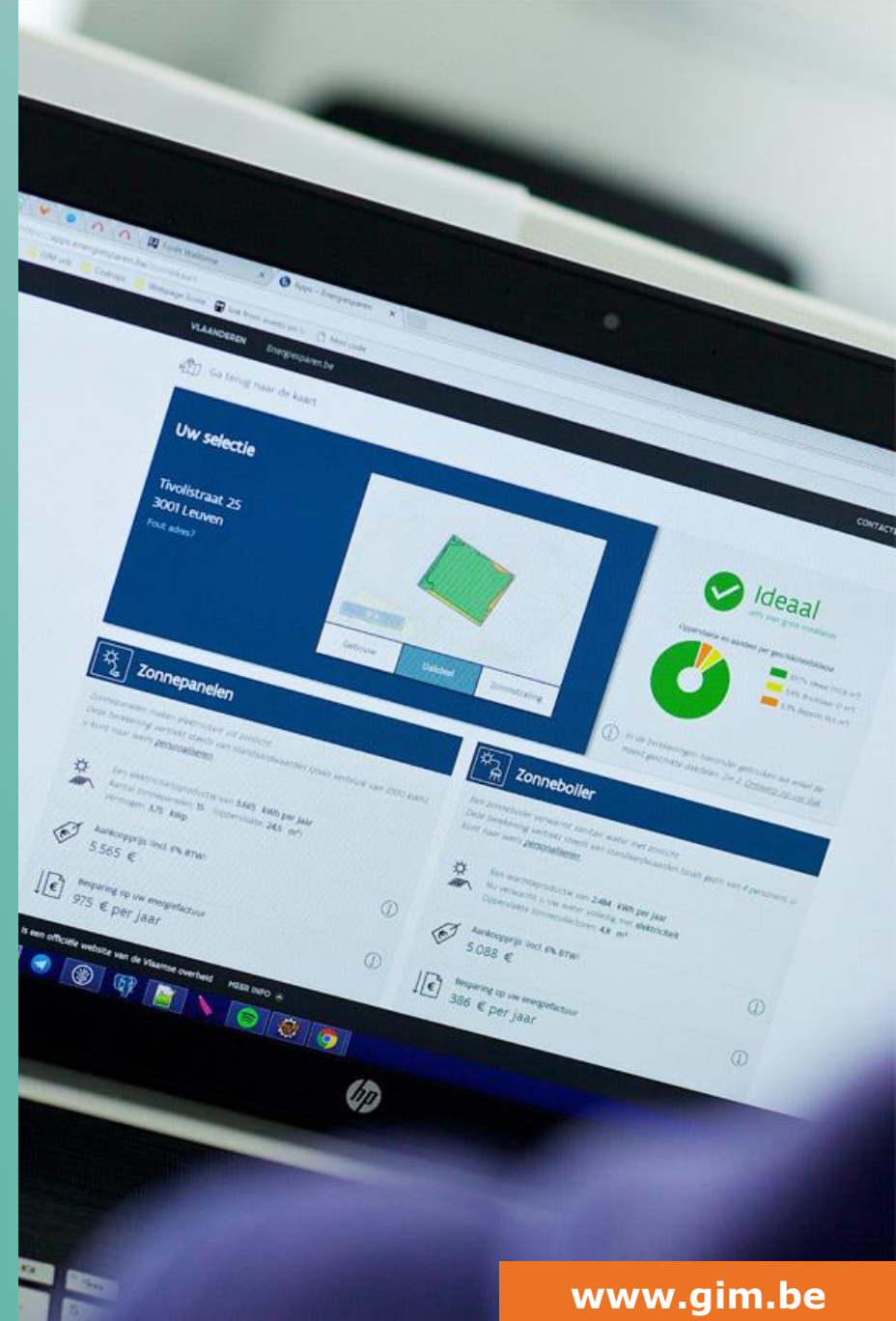
- Enable **displaying layers** according to a configured **time range**
- Steps to configure:
 1. Set the **time range** by layer (in the properties: by attribute or manually)
 2. **Enable** temporal controller
 3. Choose the **time mode** (fixed or animated)
 4. **Launch** the animation (if selected)
- Can export images

Specific use of this tool within this project

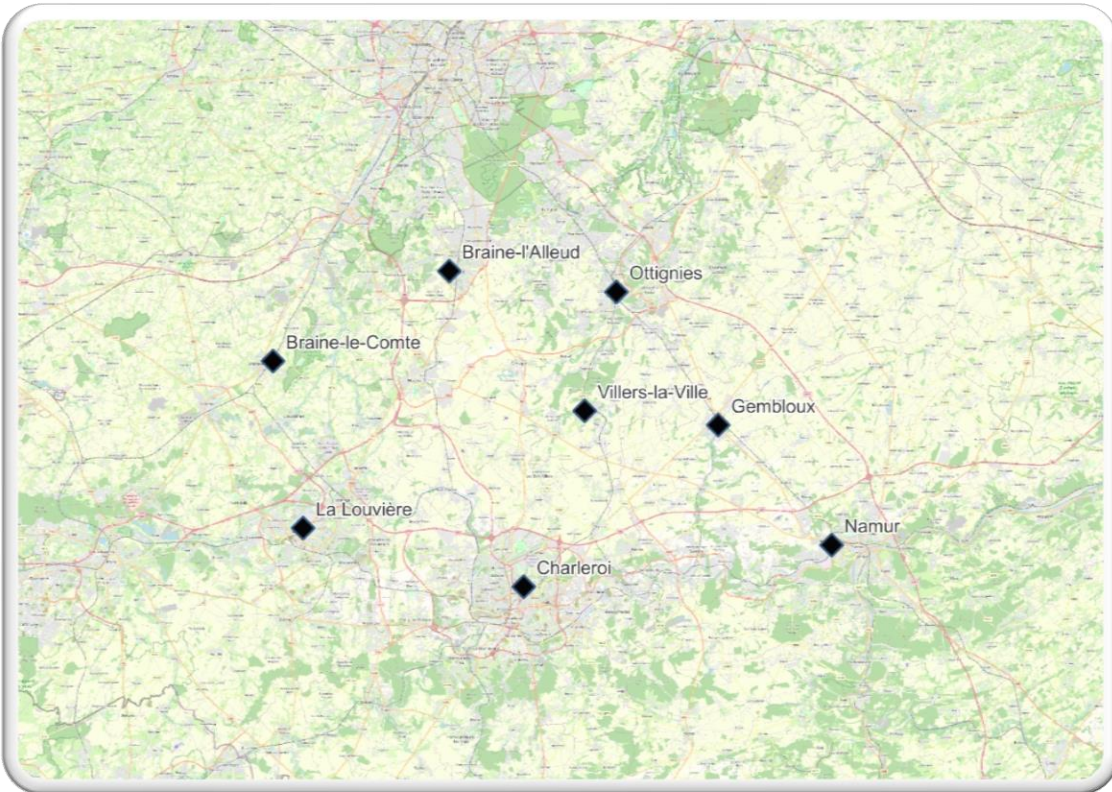


- Should be used for **evolution in time**
- Here used to **present different scenario**
 - Allow **non-QGIS user** to present different scenario quickly (no need to play with layers)
 - **Switch** a set of layer **on and off** automatically
 - Why not only use PDF ?
Allow the user to **improvise a scenario** by activating other layers during the presentation if needed

3. Step by step

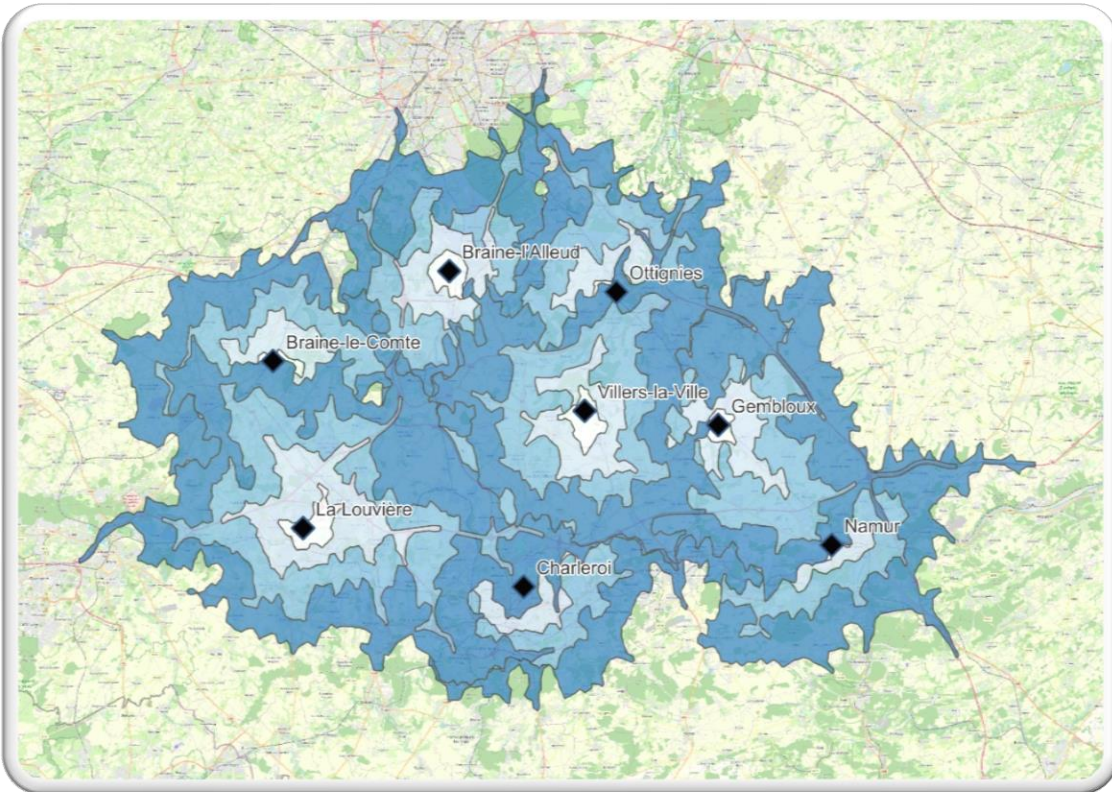


Step by step



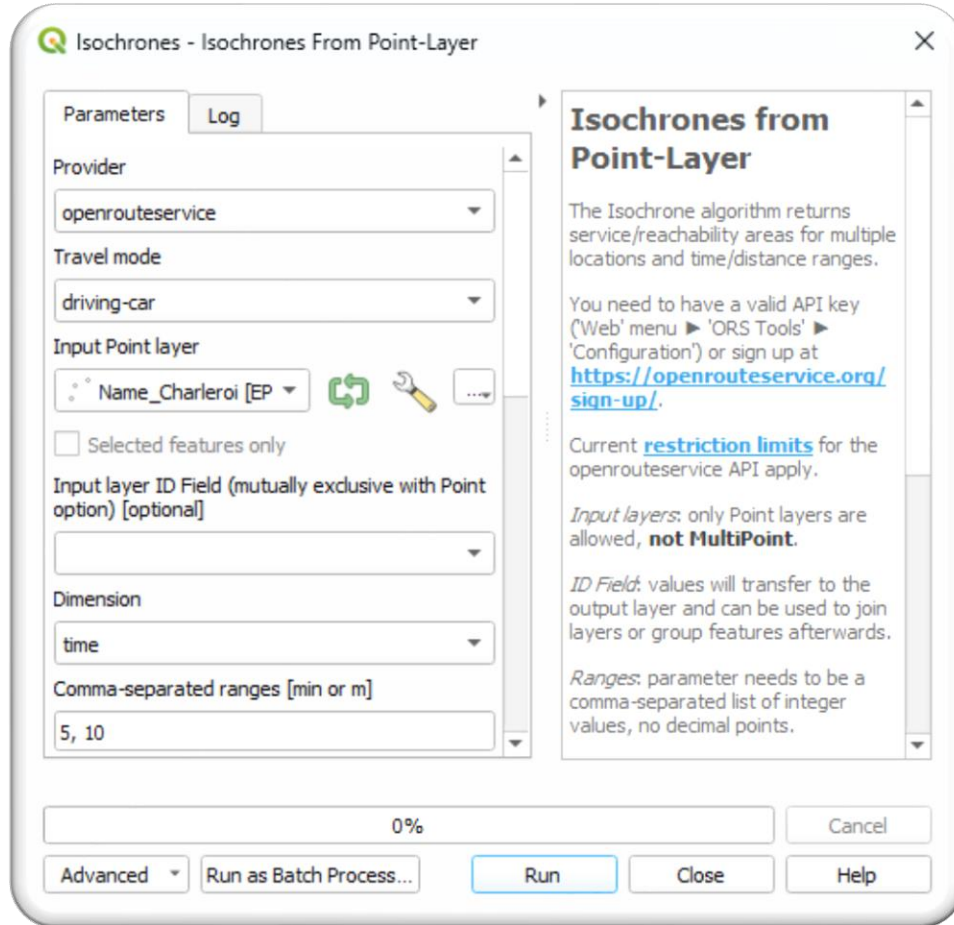
1. Creation of the point layer for service centers

Step by step



1. Creation of the point layer for service centers
2. Creation of the isochrones (**with ORS Plugin**)

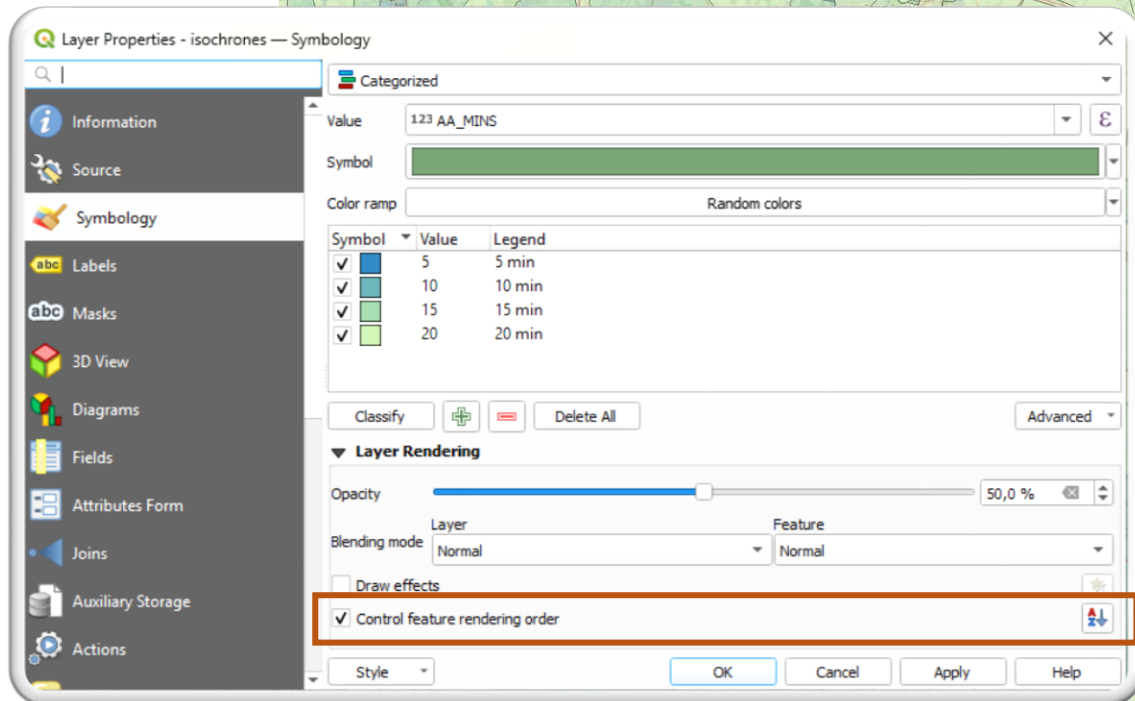
Creation of the isochrones



Parameters:

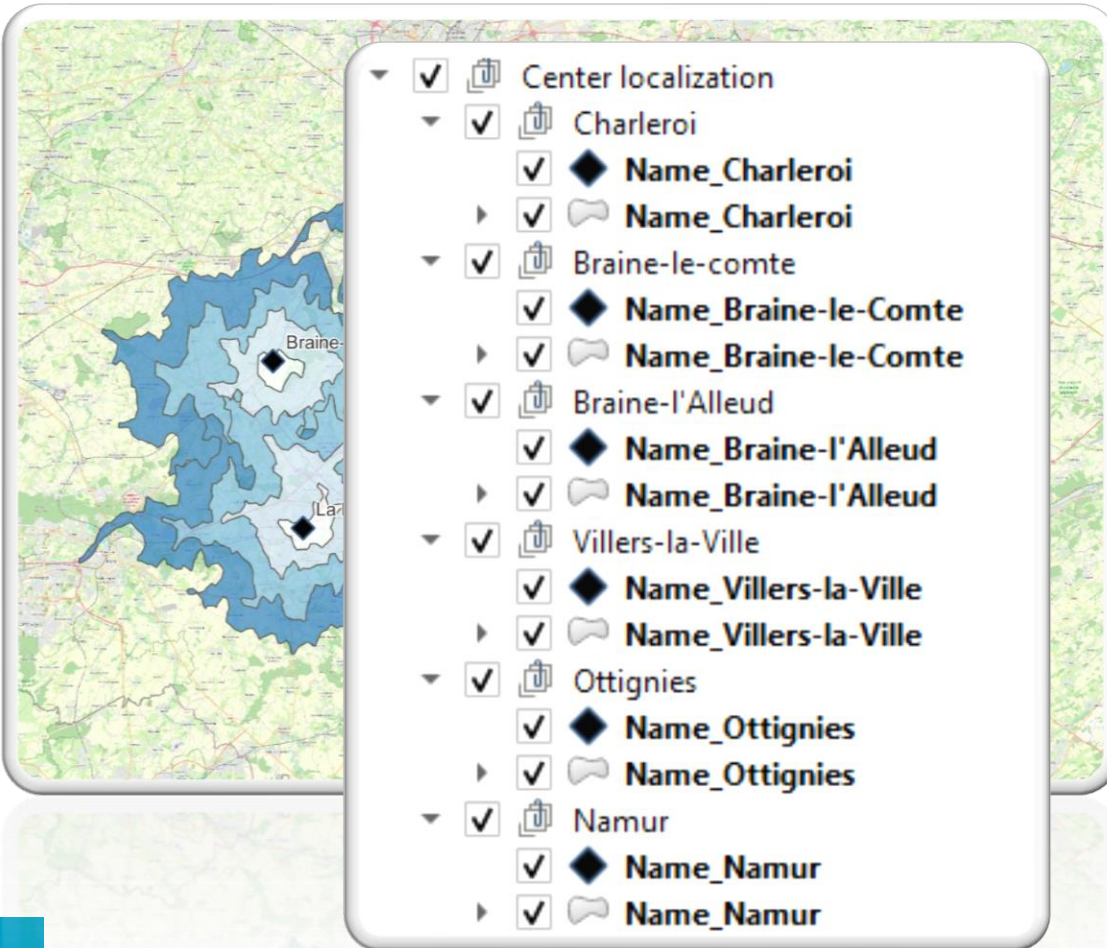
- Isochrones from layer
- Travel mode: *Driving by car*
- Input point layer: *Service centers*
- Input layer ID: *Name*
- Dimension: *Time*
- Ranges: *5, 10, 15, 20 min*
(always better to avoid more time ranges at this step)
- Location type: *Start*

Good to know



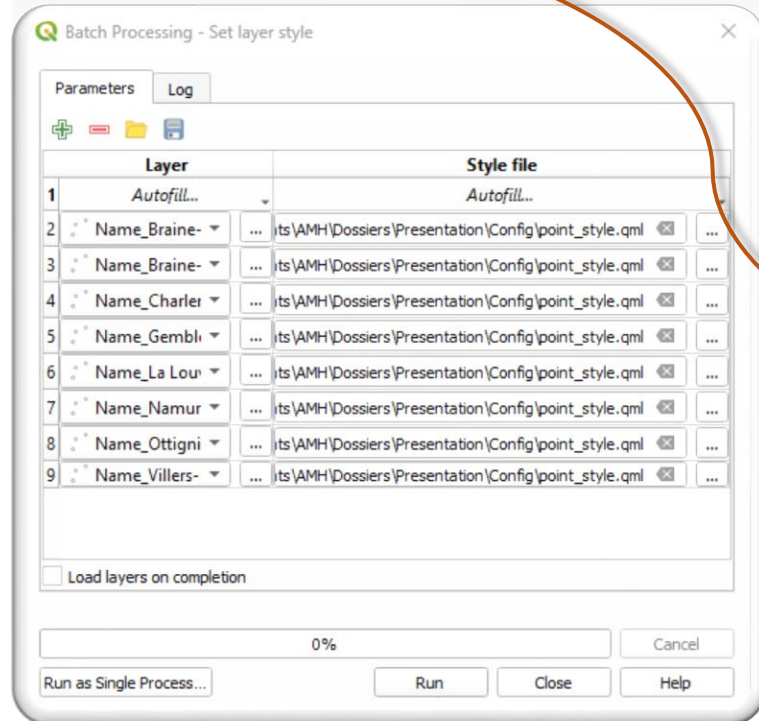
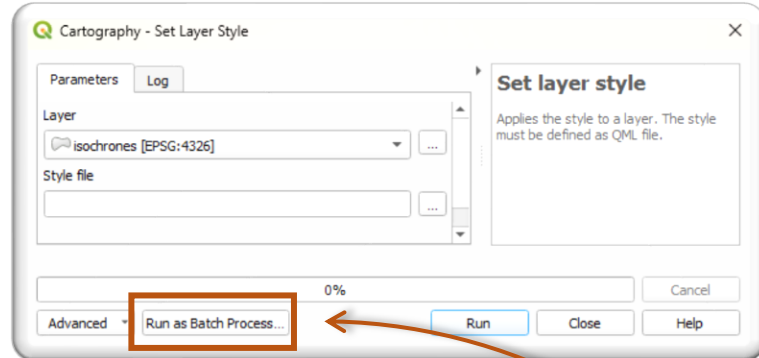
- Isochrones generated by ORS plugin are not **ordered correctly**
- To display the smallest isochrones on top
→ Use the « **Control feature rendering order** » of the symbology panel

Step by step



1. Creation of the point layer for service centers
2. Creation of the isochrones (**with ORS Plugin**)
3. Split by center localization (**Tool split vector layer**) → *One center : one layer*

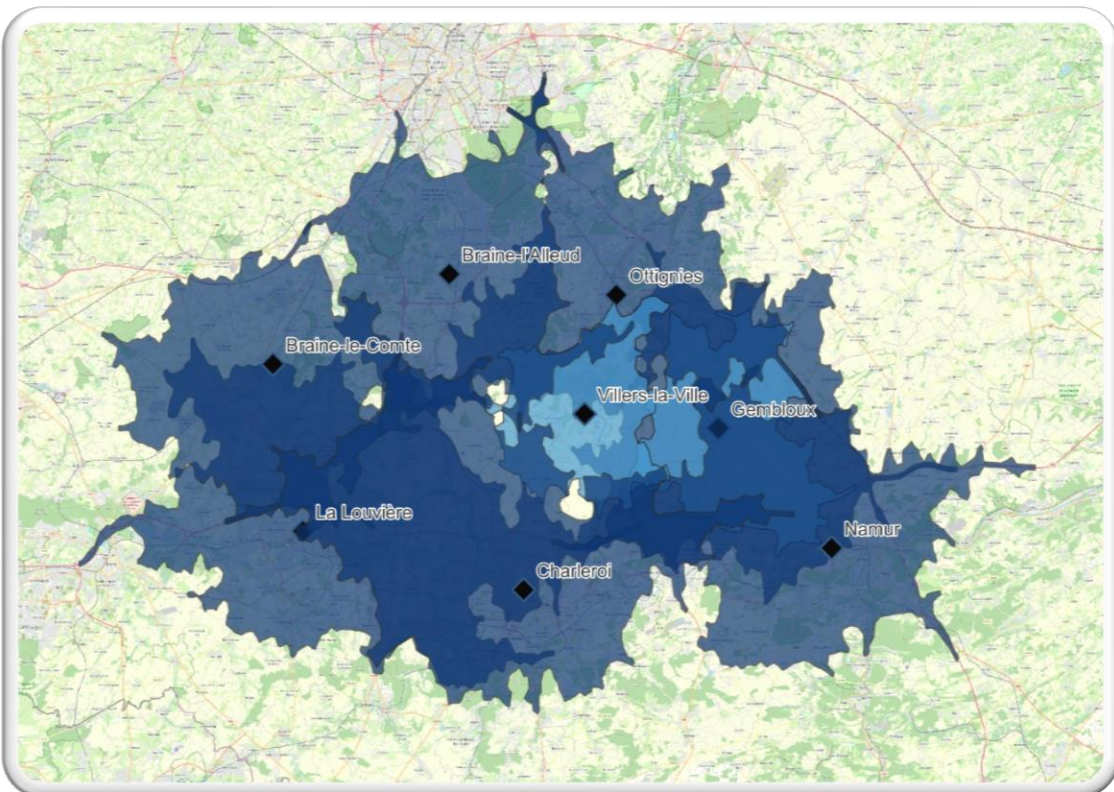
Good to know



Set layer style – **Batch processing**:

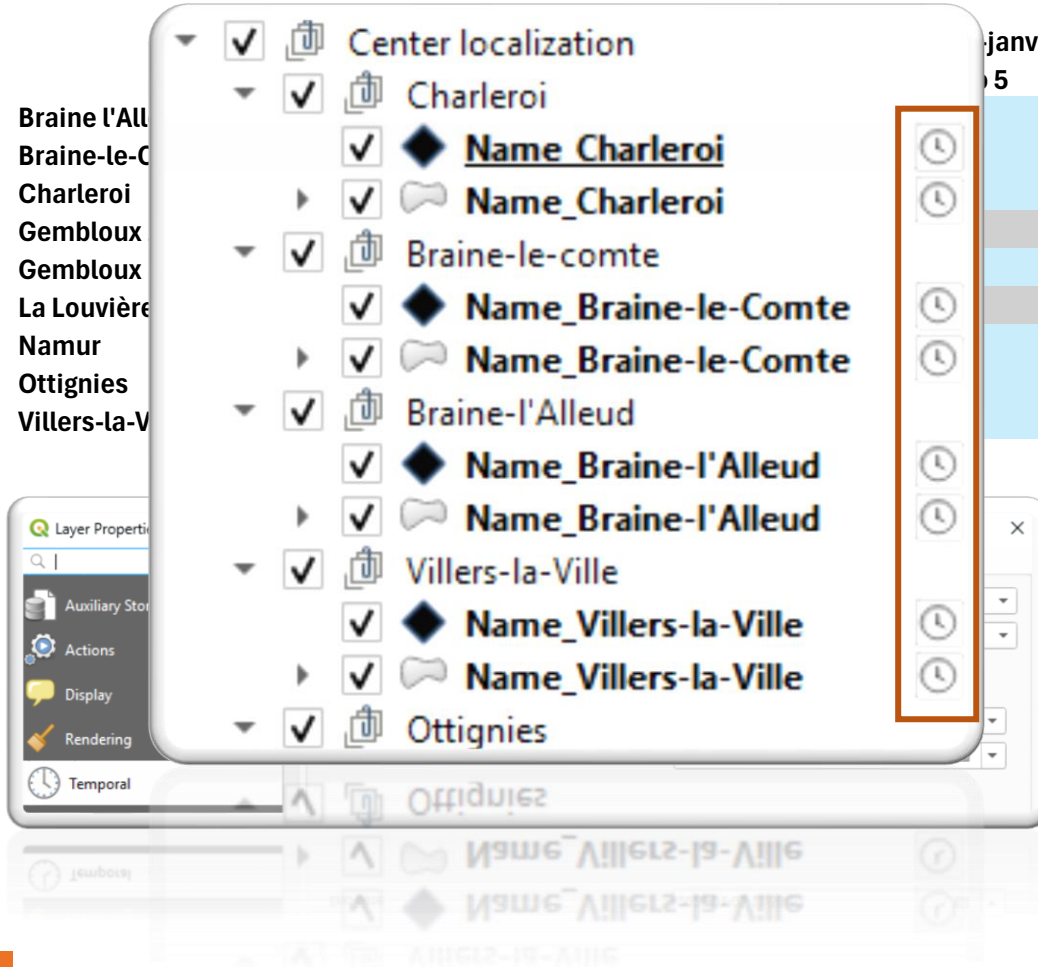
- For the temporal control, need to split data by layer
- To quickly change the style of all layers
 - Save a **template of the style** (.qml)
 - Tool: « **Set layer style** »
 - **Batch processing option** to apply the style to all layer at a time

Step by step



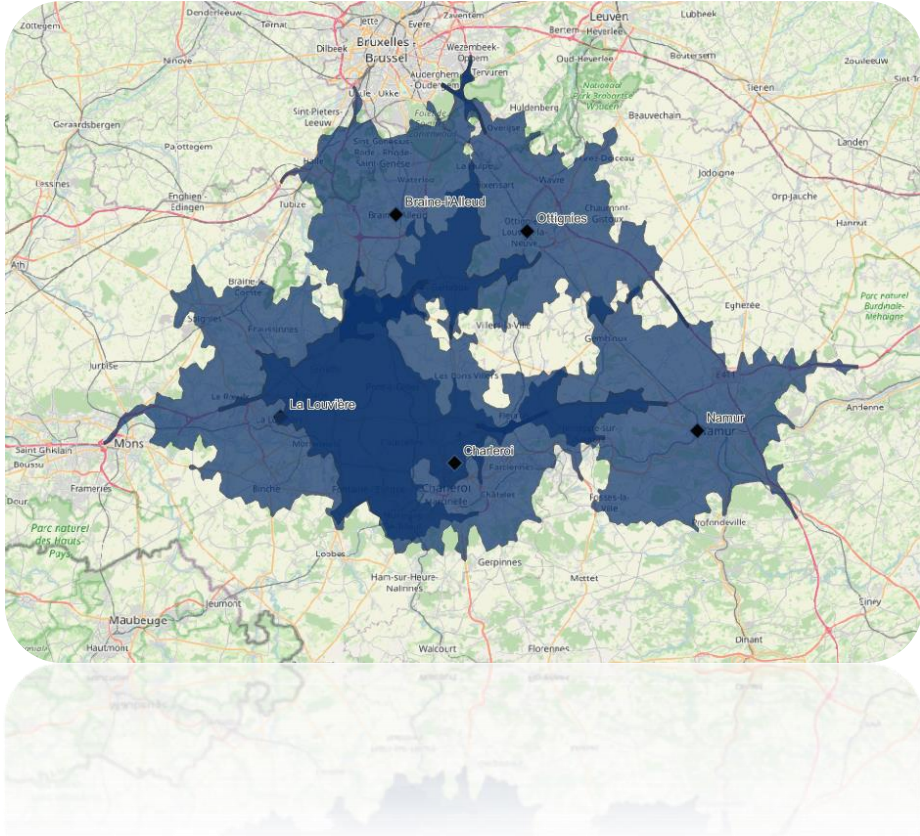
1. **Creation of the point layer** for service centers
2. Creation of the isochrones (**with ORS Plugin**)
3. Split by center localization (**Tool split vector layer**) → *One center : one layer*
4. Choice of the **time range by center** (according to the center type)
5. Establishment of the **temporal propriety**

Apply scenario to temporal control



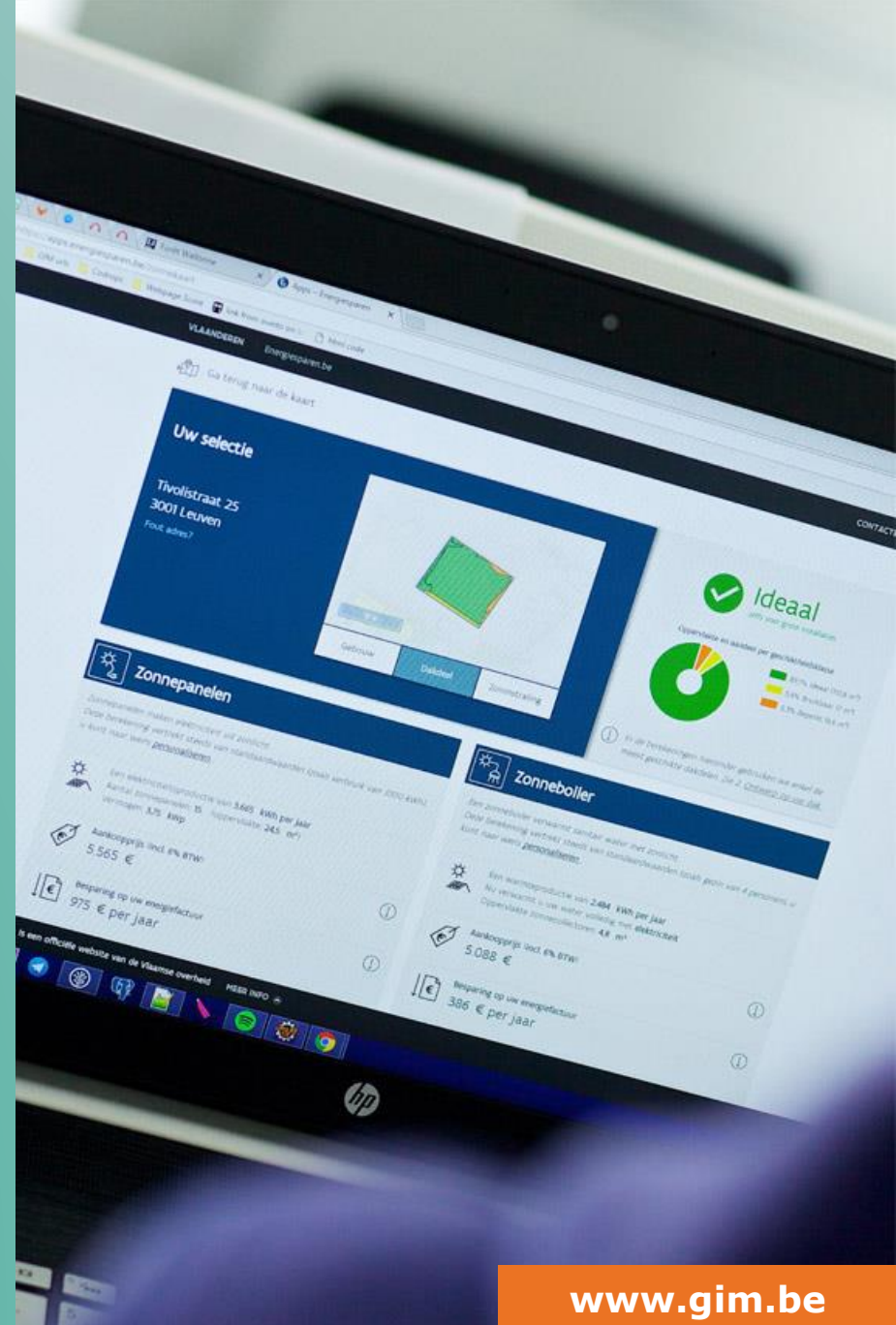
- Scenario **fixed** previously in a table
- To use the temporal tool here
 → **one day : one scenario**
- Set in the **properties** (can be set with attribute)

Step by step



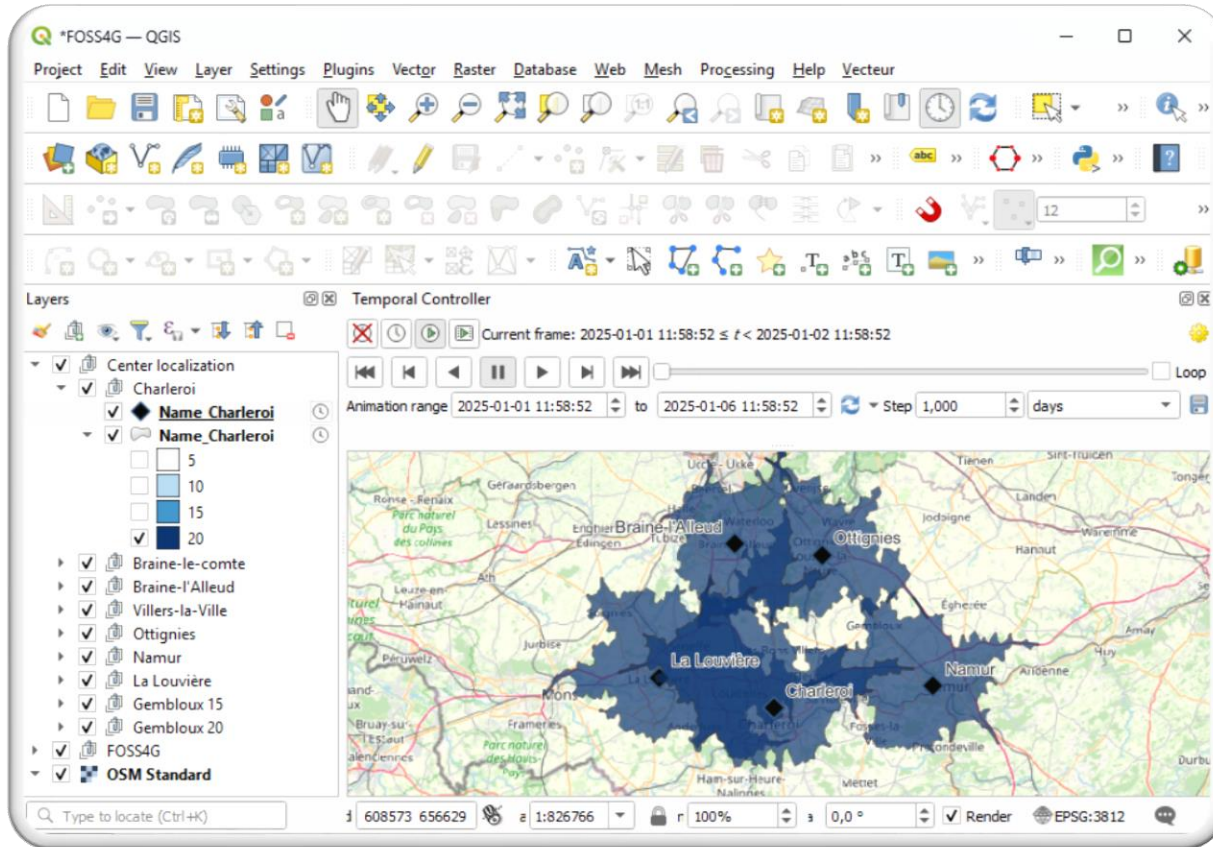
1. Creation of the point layer for service centers
2. Creation of the isochrones (**with ORS Plugin**)
3. Split by center localization (**Tool split vector layer**) → *One center : one layer*
4. Establishment of the **temporal propriety**
5. Choice of the **time range by center** (according to the center type)
6. Configuration of the temporal **dynamic control**

4. Conclusion



Conclusion

USE OF QGIS AND ORS



- **Easy to create** – No API knowledge needed
- Qgis project created: **decision-making tool**
- Offers a **customisable** presentation → Ability to **switch between scenario** during the discussion and create new ones
- Qgis project **final output** for the presentation

Any questions?

